 not the main genus to be described, genera related to Biotech Ltd., 1
India; catalog No. D5525-01) was used for Bacillus within the family Bacillaceae, such as DNA extraction. The resulted DNA was finally stored at Geobacillus and Anoxybacillus, were described as the -20°C for further analysis. main bacterial genera [2]. Similar studies in other parts of the world have Amplicon Sequencing: Amplicon sequencing using next revealed almost the same conclusion, i.e., Bacillus and generation technology (bTEFAP®) was used to analyze Bacillus-related genera are the main genera in hot springs. the water samples as described previously [14]. Briefly, For instance, a total of 79 bacterial isolates dominated by the 16S universal eubacterial primers 515F the genus Bacillus were obtained from four hot springs in (GTGCCAGCMGCCGCGGTAA) and 806R Morocco [10]. In another study, a collection of 161 strains (GGACTACHVGGGTWTCTAAT) were used to evaluate of thermophilic Bacillus were also isolated from Tunisian the species richness and abundance of each sample hot springs [11]. Additionally, different Indian hot springs including the genus Bacillus on the Illumina MiSeq with were studied and several strains belonging to the genera methods via the bTEFAP® DNA analysis service. The Bacillus and Geobacillus were isolated [12, 13]. PCR using Hot Star Taq plus Master Mix Kit (Qiagen, Recently, we have applied a culture-independent Valencia, CA) was done as described previously [14]. approach to reveal the microbial diversity in Jordanian hot Samples were sequenced utilizing the Illumina MiSeq springs represented by Ma'in and Afra hot springs [14]. chemistry following manufacturer’s protocols. The Q25 As a spinoff of this study, important data about the sequence data derived from the sequencing process were diversity of certain genera became available. In this processed using a proprietary analysis pipeline current study, we aimed primarily to review the abundance (www.mrdnalab.com, MR DNA, Shallowater, TX). of the genus Bacillus in Jordanian hot springs (Mainly Ma'in and Afra springs) based on these data generated by Data Analysis: Visualization and analysis of results was the Illumina MiSeq which was not utilized before for this done using Krona software. Krona software allows purpose. Secondly, the species composition of the genus hierarchical data to be explored with zooming, Bacillus would be determined because most studies did multi-layered pie charts. Microsoft Excel 2010 was also not identify the thermophilic/thermotolerant Bacillus used to generate the abundance charts. isolates to the species level. MATERIALS AND METHODS The new findings in this study were generated from unpublished metagenomic data emerged from a recent published work done by our group on samples obtained from Ma'in and Afra hot springs [14]. Sampling of Hot Springs Water: Water samples were collected from four Jordanian hot springs; three samples were collected from Ma'in hot springs (HS1, HS2 and HS3) and one sample from Afra hot springs (HS4) as mentioned before [14]. Sample collection was carried out in June, 2015 as mentioned previously by our group and the in situ temperature of water ranged between 38 to 59°C whereas the pH values rangedbetween7.4 to 8.4 [14]. DNA Isolation: DNA was isolated as described before [14], briefly, water samples were filtered under vacuum using 0.2 m- nitrocellulose membranes. The nitrocellulose membranes containing the trapped cells were cut into pieces and transferred to sterile 50 mL centrifuge tube. The E.Z.N.A Water DNA kit (Omega ® RESULTS Abundance of the Genus Bacillus: The abundance of the genus Bacillus within the family Bacillaceae was determined relative to different taxonomic levels. The genus Bacillus was found predominating the family Bacillaceae in three samples out of four (HS2, 84%; HS3, 77% and HS4, 61%; Table 1). However, in sample HS1, the genus Bacillus was found to be the second most abundant genus (22%; Table 1). The abundance of Bacillus was also calculated relative to higher taxa. The genus was dominating at the order Bacillales and class Bacilli) only in case of HS2 (64%). However, the abundance was lower in HS1 (18%), HS3 (47%) and HS4 (39%; Table 1). At the level of the phylum Firmicutes, the genus constituted 7% in HS1, 41% in HS2, 31% in HS3 and 33% in HS4 (Table 1). Relative to the domain of Bacteria, the abundance of the genus was found to be very low: 2% (in HS1), 3% (in HS2), 0.5% (in HS3) and 1% (in HS4; Table 1). Because Bacillus was not the dominant genus within the family Bacillaceae in HS1, further investigation was carried out to clarify the relative abundance of genera other than Bacillus within the family